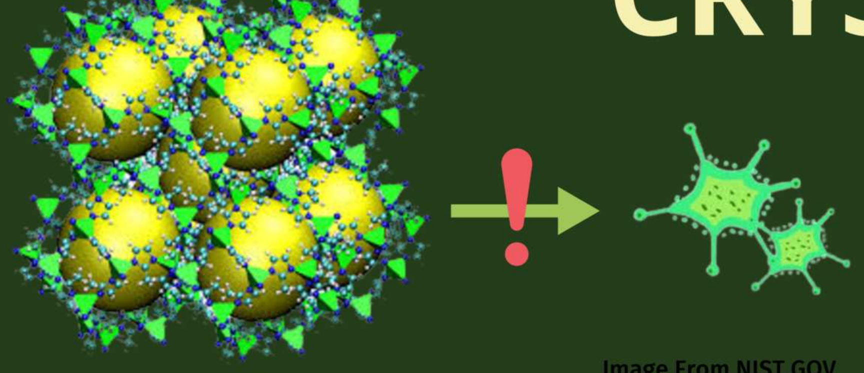
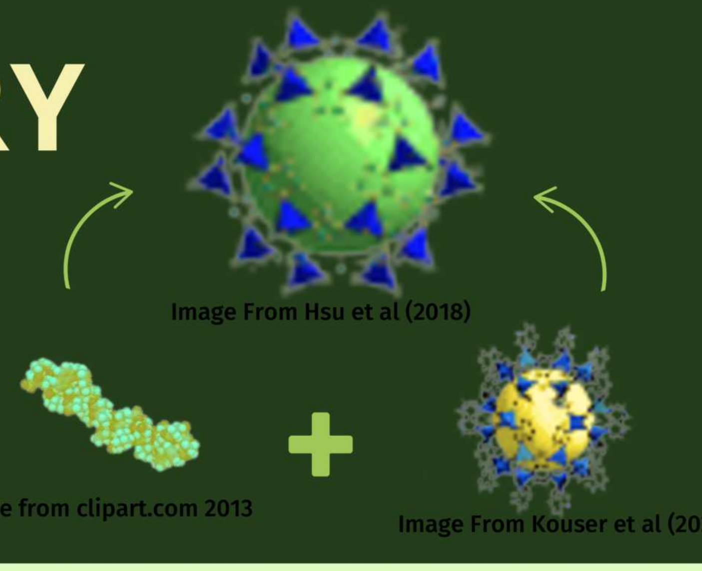


CRYSTALCARE: OPTIMIZED NANOCRYSTALS FOR DRUG DELIVERY

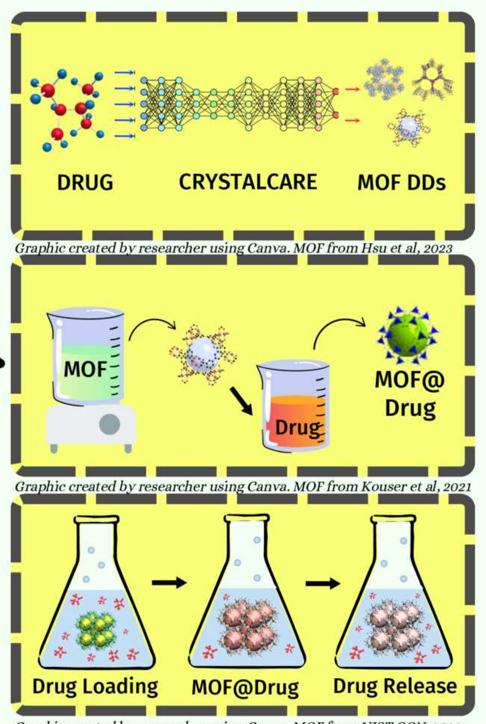
Discovering Optimized Nanocrystal Framework Drug Delivery Systems For Diversified Delivery Applications

ENBM | 514-X-12



BRIEF OVERVIEW

This year's research aimed to engineer a graph attention-based computational architecture to identify & optimize Metal-Organic Frameworks as drug delivery systems. Top targets were synthesized via a UARM method and experimentally compared to standard-of-care drugs. The identified MOFs exhibited a significant (p<0.01) increase in drug loading when compared to leading pharmaceutical products. The CRYSTALCARE architecture presents a striking framework for pharmacological innovation.



Graphic created by researcher using Canva. MOF from Hsu et al, 2023

Graphic created by researcher using Canva. MOF from Kouser et al, 2021

Graphic created by researcher using Canva. MOF from NIST.GOV, 2022

INTRODUCTION

DRUG DELIVERY

Over 90% of viable drugs fail the FDA process, with over 50% of conventional drugs failing to reach their intended site of action due to rapid degradation and poor bioavailability. New approaches are required to overcome biological barriers and ensure that therapeutic agents are released precisely at the disease site. Developing novel, biocompatible carrier systems that provide targeted, controlled-release drug loading is an crucial area for improving clinical outcomes in chronic diseases and infections.

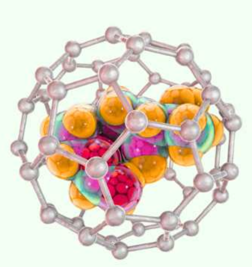


Image From Dierks (et al, 2017)

METAL ORGANIC FRAMEWORKS (MOFs)

MOFs are porous materials constructed from metal ions and organic linkers, known for exceptional surface area. They are utilized for high-capacity adsorption, antibacterial, and drug delivery applications. Further research on stabilizing these structures for therapeutic and biomedical use is needed.

HIGH SURFACE AREA
Maximizes high drug loading (7,000 m^2/g!) and antibacterial effectiveness

TUNABLE CHEMISTRY
Customizable structures enable selective antibacterial activity

DRUG DELIVERY
Enhances high drug stability, targeted release, and antibacterial effects

OBJECTIVES

This research hopes to redefine the current methods of drug delivery design. Instead of aiming to arbitrarily select drug delivery systems for experimental verification, this research aims to design an optimization architecture that identifies MOF candidates with high potential as a drug delivery system for an inputted drug, ultimately increasing ADME scores for drug use.

NOVEL ARCHITECTURE
Use attention based mechanisms to learn drug-MOF pairs.

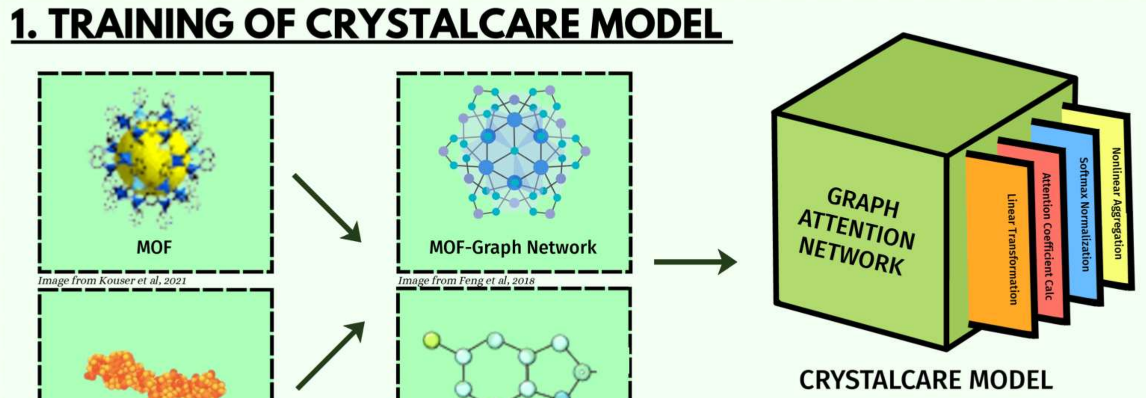
TRANSPARENT RESULTS
Highlight valuable drug delivery MOF sites for researchers to analyze.

EXPERIMENTAL ANALYSIS
Experimentally synthesize and verify MOF DDs using drugs.

In addition, this research hopes to present new insights into drug loading and delivery in the field of skin infections, a field that rarely has innovation in drug optimization and application. This research can hopefully provide insights into drug loading and possibility of new treatment mechanisms that can be applied to new applications, such as other biomedical sciences.

METHODOLOGY

1. TRAINING OF CRYSTALCARE MODEL

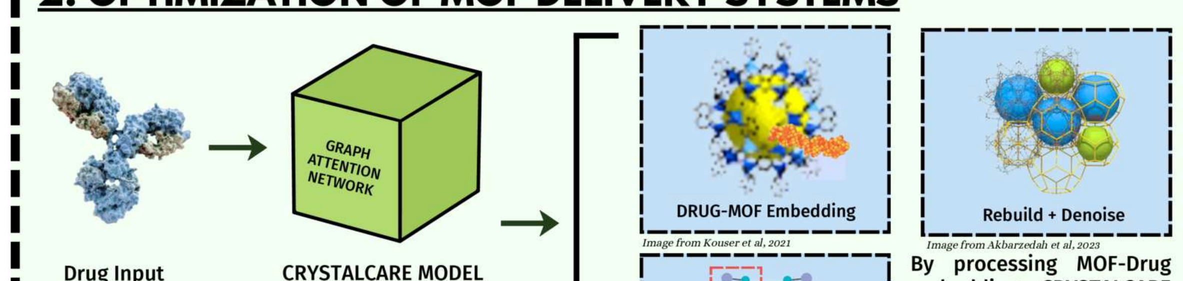


CRYSTALCARE MODEL

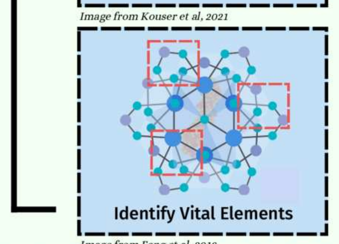
MOF and drug molecules were transformed to graph networks, with atom-nodes & bond-edges. Features were atom-encoded.

Graphics created by researcher using Canva

2. OPTIMIZATION OF MOF DELIVERY SYSTEMS



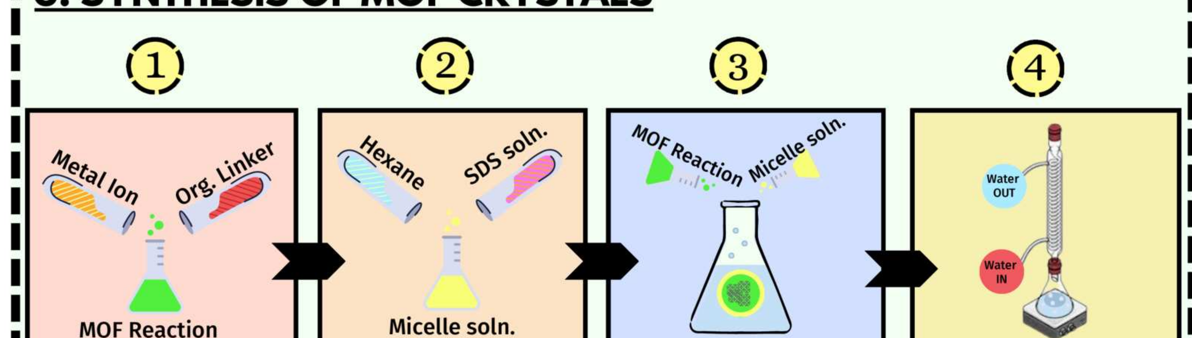
After model training, new drug samples were provided to identify optimized MOFs through screening (experimental literature MOFs) and rebuilding (hypothetical MOFs). Experimental MOFs were chosen for synthesis.



By processing MOF-Drug embeddings, CRYSTALCARE can effectively relearn important structural MOF components conducive to high drug delivery scores. Deletion and rebuilding allows for ablation model optimization of MOFs.

Graphics created by researcher using Canva

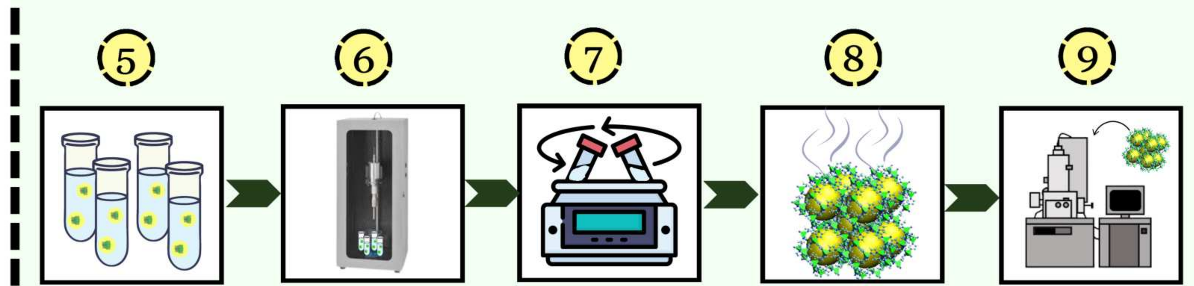
3. SYNTHESIS OF MOF CRYSTALS



1. **MOF Reaction** — MOF solution is formed through reaction of a metal ion solution and an organic linker.
2. **Micelle soln.** — A reverse micelle formation is formed through reaction of hexane and SDS soln.
3. The MOF synthesis is began by reacting the micelles solution with the MOF solution.
4. MOF solution is reacted in reflux for 1 hour at 85 C. Evaporation of hexane is prevented.

Graphics created by researcher using Canva

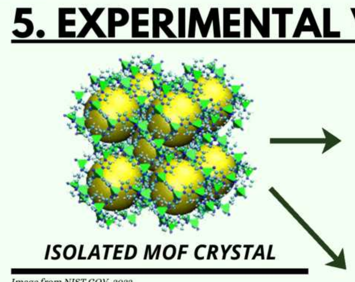
4. RELEASE AND ISOLATION OF MOF CRYSTALS



The MOFs are released and isolated from the solution through ultrasonic treatment. The sonication releases the trapped MOFs from inside the reverse micelles, leading to micelle byproduct and MOF solid. Afterwards, the solution is centrifuged to isolate the MOFs in a solid form. MOFs are formed to crystal by allowing to evaporate and dry in fume hood.

Graphics created by researcher using Canva

5. EXPERIMENTAL VERIFICATION OF DRUG DELIVERY

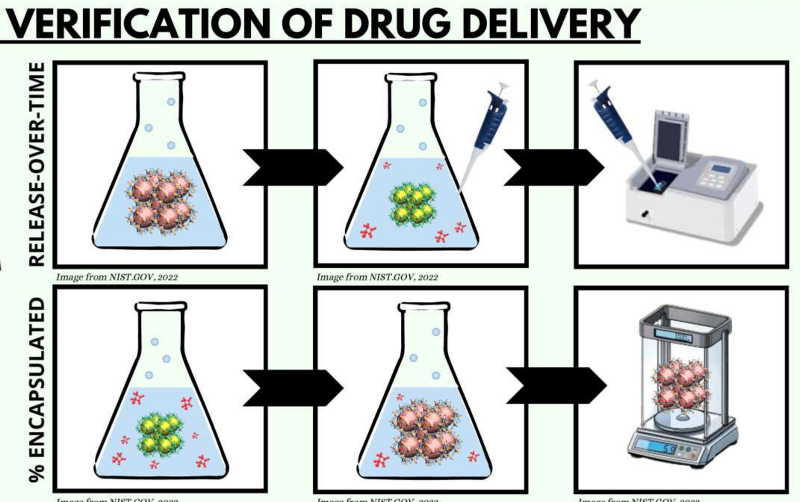


ISOLATED MOF CRYSTAL

Image from NIST.GOV, 2022

RELEASE-OVER-TIME:
Supernatant was sampled at specific time intervals with UV-VIS to quantify drug amount.

PERCENT ENCAPSULATED:
MOF crystals were weighed, then loaded with 10 mmol of drug. The change in mass was calculated.



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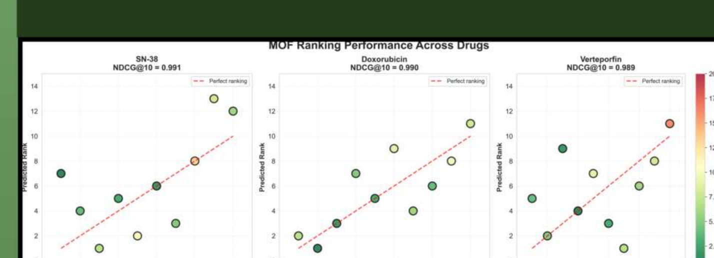
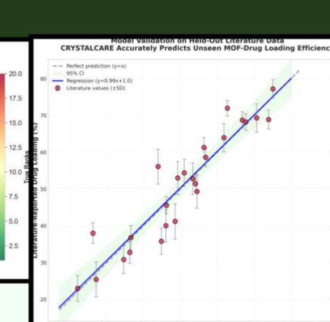
OVERVIEW OF METHODOLOGY

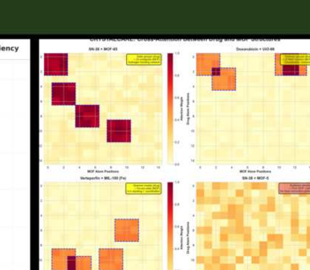
In this project, an attention-based graph neural network was trained by transforming MOF & drug molecules into graphical representations and training. The model was used to identify optimized MOF delivery systems for experimental validation on three selected drugs. The three identified MOFs were synthesized via a UARM (ultrasonic-assisted reverse micelle) process and quantified through FTIR and SEM. MOFs were loaded with their respective drugs post-synthesis, whereby a drug-loading assay and release-over-time assay were completed. Drug release was quantified with UV-VIS spectroscopy curves.

RESULTS AND SYNTHESIS ANALYSIS

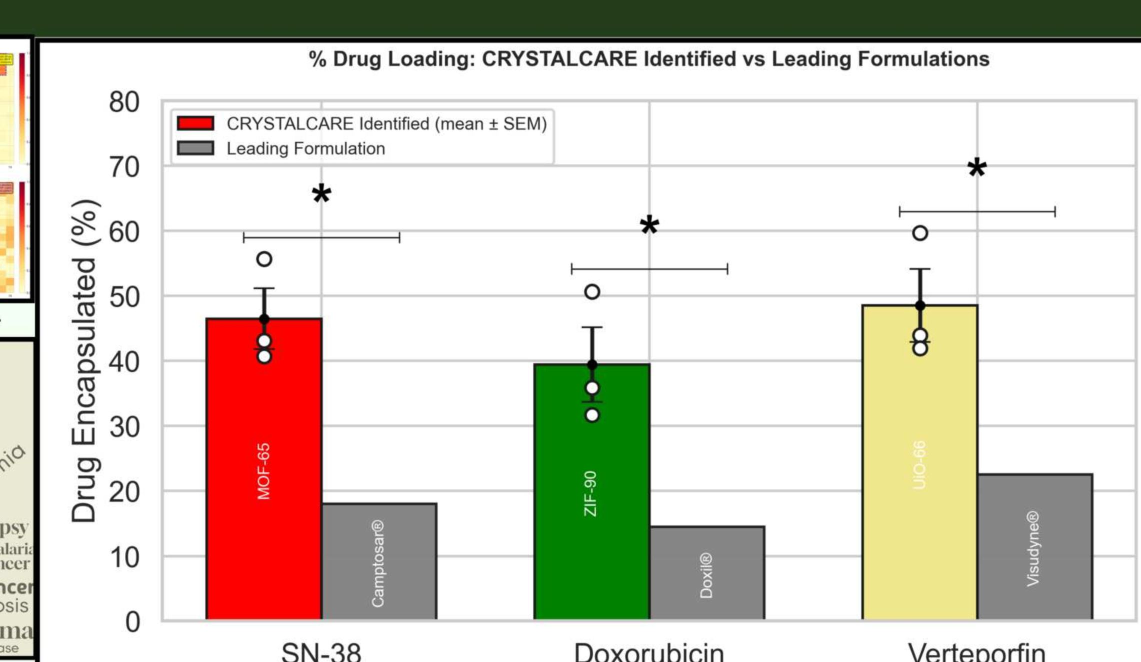


Normalized Discounted Cumulative Gain at 10 rankings for experimental drugs. CRYSTALCARE was effective at learning valuable ADME indicators (r=0.93).
Graphs made by researcher in Python

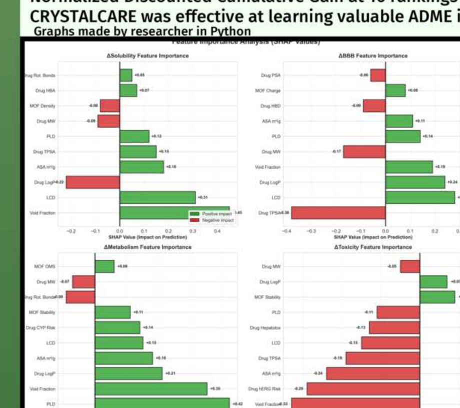




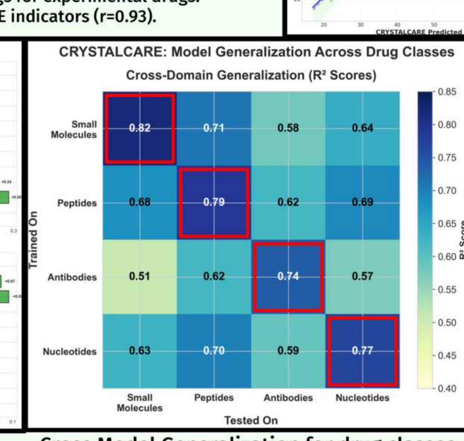
MOF-Atom Attn. heatmaps.
Graphs made by researcher in Python



CRYSTALCARE-Identified MOFs improved drug loading over literature scores of standard-of-care formulations. Literature values were compared by loading DDS, centrifuging, and replacing supernatant for mass change.
Control values were obtained from Feng et al, 2014 (Camptosar), Russell et al, 2017 (Doxil), Siriwardane et al, 2021 (Visudyne). Graph was created by researcher using Python.

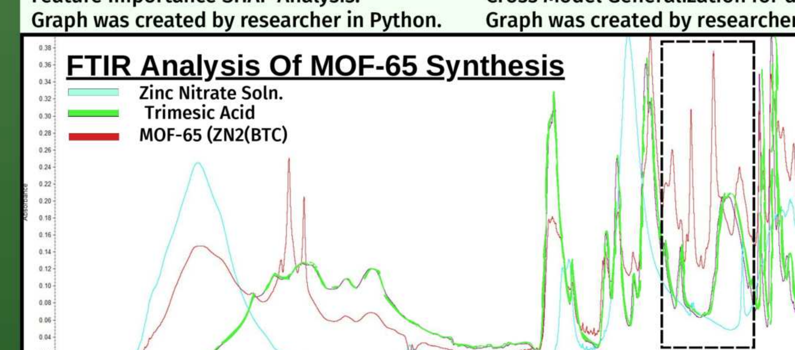


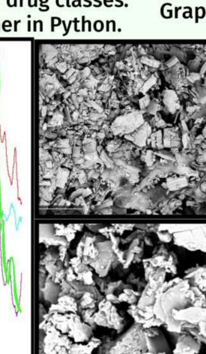
Feature Importance SHAP Analysis.
Graph was created by researcher in Python.



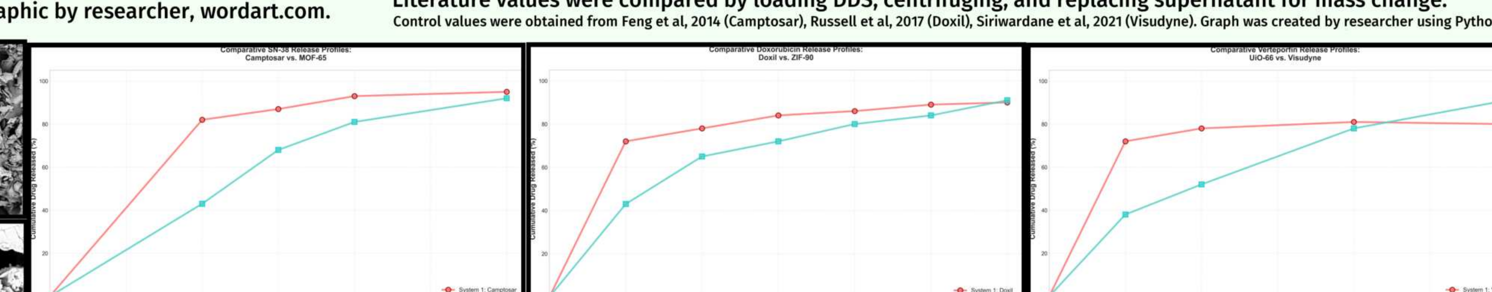
Cross Model Generalization for drug classes.
Graph was created by researcher in Python.

Wordmap of diseases that CRYSTALCARE has identified MOF delivery systems for. Size represents model confidence. Graphic by researcher, wordart.com.





FTIR for MOF-65 shows new bonds in the C-C region → trimesic framework bonds. | SEM→nanocrystal features
FTIR spectra obtained by researcher using FTIR-4390. | SEM images obtained by researcher using ZEISS EVO SEM

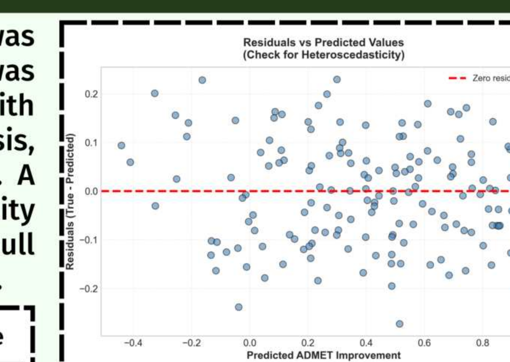


A key concern of many drug candidates is effective pharmacological drug release timescales. Drugs with poor release scales often provide their drug in large dosages in a short time scale, preventing sustained control and diminishing viability and patient compliance. CRYSTALCARE MOFs were compared experimentally to Standard-of-Care drugs by loading DDs and sampling supernatant with UV-VIS. Supernatant was replaced and maintained at pH=7.4 and equivalent temperatures to reported controls. MOFs showed more sustained drug release.
Control values were obtained from Feng et al, 2014 (Camptosar), Russell et al, 2017 (Doxil), Siriwardane et al, 2021 (Visudyne). Graph was created by researcher using Python.

*All Graphs made by researcher using Python: Matplotlib, Seaborn, MS VSCode

STATISTICAL ANALYSIS

A Breusch-Pagan test for homoscedasticity was run to analyze if residual variation was dependent on the independent variable. With p=0.87, we fail to reject the null hypothesis, suggesting that residuals are independent. A Shapiro-Wilk test was run to analyze normality of residuals; with p<0.01, we reject the null hypothesis; residuals are abnormally spread.



Graph made by researcher using Python

MOF DDs	Mean	SOC value	U	p value
MOF-65	47.453	18.756	0.00	0.02472
ZIF-90	39.928	14.392	0.01	0.02676
UiO-66	48.628	23.362	0.04	0.03245

Table made by researcher using Canva

Mann-Whitney U was used for drug loading significance in difference between MOFs and Standard-Of-Care. Null hypothesis was rejected.

CONCLUSIONS

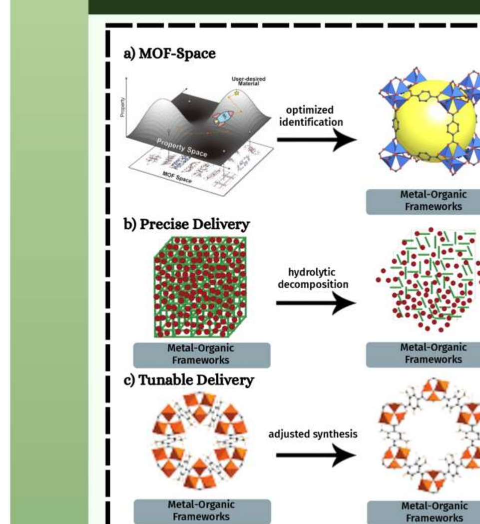


Image Taken From Matusiak et al (2023)

MODEL VIABILITY

The CRYSTALCARE model learned drug delivery patterns of MOF molecules. This was validated using weight permutations and comparisons. CRYSTALCARE achieved 93% accuracy. The model showed signs of generalizability, highlighting attention between MOF-node clusters at higher-order scales. This model can empower drugs through identification of optimal drug delivery systems, increasing the likelihood of drugs reaching FDA approval

MOF DRUG DELIVERY

This project synthesized the top three identified MOFs from the CRYSTALCARE model. These MOFs were quantified using FTIR and SEM. MOFs improved delivery by drug loading over standard-of-care drugs (p<0.01). MOF drug delivery systems showcase significant promise for future clinical research and testing.

MODEL NOVELTY
CRYSTALCARE used a attention head to learn local and global patterns

DRUG DELIVERY
Optimized MOFs showed high delivery potential through high drug loading.

POTENTIAL RESEARCH
CRYSTALCARE-identified MOFs can encapsulate drugs before testing.

FUTURE GOALS

1. The CRYSTALCARE model can be used to identify possible drug delivery systems for all failed drugs in the FDA system's history, empowering millions of new treatments.
2. MOF Drug delivery systems identified by CRYSTALCARE must be subjected to rigorous in vivo testing in both cell and mammalian models to ensure delivery predictions.
3. Further research into the new designs of the hypothetical, optimal MOFs provided by CRYSTALCARE could realign materials science researchers with chemical design goals.

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